

Technology #141

Implementing a cost-effective battery management system with enhanced monitoring and control capabilities that can detect abnormal charging patterns, prevent overcharging, and isolate cells showing signs of degradation

Engineering Technology-Transfer Dossier

CONFIDENTIAL

Technology Maturity: ENGINEERING_DEFINITION

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Not physically validated. Not transfer-ready.

BUYER DECISION PAGE

WHAT IS IT?

Implementing a cost-effective battery management system with enhanced monitoring and control capabilities that can detect abnormal charging patterns, prevent overcharging, and isolate cells showing signs of degradation

WHY DOES IT MATTER?

Documented recurring problem 'lithium-ion consumer-product battery fire': 5 of 50 CPSC recalls in the retrieved window involve batteries/chargers/power electronics. Recalled products: 'Cooluli Recalls 10-Liter and 15-Liter Minifridges Due to Fire and Burn Hazards'; 'Shengsen Prank Sound Makers Recalled Due to Risk of Serious Injury or Death from Battery Ingestion'; 'A2batt Recalls EEMB Lithium Coin Battery Chargers Due to Risk of Serious Injury or Death from Batter'. LIMITS STAMPED INLINE: a recall is an acknowledged product hazard, NOT an incidence rate; injury lists are associated complaints, not a census (Art. XXI.5 analog).

WHAT IS ACTUALLY ESTABLISHED?

Engineering definition with 3 governing equations, 12 design inputs, 6 failure modes. Computational model exists. No physical prototype.

WHAT IS NOT ESTABLISHED?

Physical validation, manufacturing process, regulatory pathway, IP ownership.

WHY MIGHT A BUYER CARE?

Opportunity in mechanical_structural. 8-step build plan with specific test articles and acceptance criteria. Target buyer: domain-appropriate technology buyer (to be identified by portfolio owner).

WHAT DOES THE BUYER GET?

Buyer receives: canonical invention specification (with epistemic classes).; Buyer receives: engineering specification (this document, all blocks classed).; Buyer receives: design input/output graph with explicit IDs.; Buyer receives: engineering reasoning chains (claim->principle->model->input->assumption->output->failure->verification).; Buyer receives: killer-experiment definition and its EIG basis.; Buyer receives: evidence custody references and hashes.; Buyer receives: unresolved-uncertainty register.; Buyer receives: domain equation set (symbolic; sources and applicability judgments identified).

WHAT DOES THE BUYER HAVE TO BUILD?

Buyer must create: actual geometry and detailed design.; Buyer must create: sourced engineering parameters and acceptance thresholds.; Buyer must create: physical prototypes.; Buyer must create: all verification and validation results.; Buyer must create: manufacturing process qualification.; Buyer must create: regulatory pathway determination and submissions.

WHAT IS THE NEXT DECISIVE EXPERIMENT?

first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed)

WHAT WOULD MAKE US KILL IT?

killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.44573539382492045, EIG-per-cost 0.44573539382492045; the falsification test carries the pre-registered pass/fail before any transfer)

WHAT TRANSACTION COULD MAKE SENSE?

Sponsored Validation

What Exists Today

Existing:

- + Documented mechanism with governing equations
- + Computational model
- + 1 external evidence sources
- + Design inputs and outputs
- + Failure analysis
- + 8-step engineering build plan
- + Transfer boundary definition

Does Not Yet Exist:

- Physical prototype
- Qualified manufacturing process
- Clinical data
- Verified IP ownership
- Regulatory clearance
- Commercial product

Buyer Can Commission:

- > Specific bench experiment (see build plan)
- > Prototype fabrication
- > Materials study
- > Manufacturing feasibility study
- > Regulatory pre-submission

Development / De-risking Ladder

CURRENT STATE: ENGINEERING_DEFINITION

NEXT ENGINEERING STEP: first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed)

NEXT VALIDATION STEP: Bench prototype validation per build plan

NEXT REGULATORY STEP: Pre-submission to FDA for pathway classification

COMMERCIALIZATION GATE: Clinical validation evidence required

TRANSFER GATE: Transfer-ready when physical validation + regulatory pathway + IP verified

Engineering Investment Ladder

Next \$5K action: WP-01 - first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed) - NOT ESTABLISHED (to be quoted)

Basis: Build plan WP. Scope: physical bench/clinical experiment — NOT simulation. Uncertainty: Effort estimate, not quoted.

Next \$25K action: WP-03 - article for VF-001 - NOT ESTABLISHED (to be quoted)

Basis: Build plan WP. Scope: Comparative testing of battery packs with and without the enhanced BMS under simulated abuse conditions including overcharging, physical.... Uncertainty: Effort estimate, not quoted.

Next \$100K action: WP-05 - article for VF-003 - NOT ESTABLISHED (to be quoted)

Basis: Build plan WP. Scope: buckling/kink margin test across the duty cycle and path geometry (measured quantity: mechanical stress / fatigue endurance). Uncertainty: Effort estimate, not quoted.

Top 5 Buyer Diligence Questions

Q1: Is the mechanism physically credible for the intended application?

The governing model includes 3 equations. Key assumptions: stress concentrators multiply local stress above the nominal value — geometry factors must be applied from the actual design; superelastic members (e.g. nitinol) do not follow linear elasticity through transformation — use their measured plateau relations instead; K for catheter constraint conditions is MODEL_DERIVED; MEC-002 (Section bending stress) is applicable only under: linear-elastic material; small deflection; prismatic section.; MEC-003 (Hooke's law (linear elasticity)) is applicable only under: below proportional limit; isothermal.; MEC-001 (Euler buckling load) is applicable only under: elastic, slender column, small deflection.; Model input variables requiring sourced values before any numeric result: M (N.m), c (m), I (m⁴), E (Pa), epsilon (-), E (Pa), I (m⁴), K (-), L (m) — all UNKNOWN (no sourced value exists; Art. VIII).; Model output variables produced by the governing model: sigma (Pa), sigma (Pa), P_cr (N) — UNKNOWN until the inputs are sourced and the model is evaluated.; Critical parameters under specification (parameters with recorded units and epistemic status; no values invented): member slenderness ratio [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; yield/ultimate strength [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; applied cyclic load amplitude [P_cr, N] value_status=UNKNOWN; required run-out cycles [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; articulation radius [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; friction coefficient at guides [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN.. UNKNOWN: whether assumptions hold in vivo.

Q2: What evidence supports the core claim?

1 external evidence sources. All are external precedent, not physical validation of this specific invention.

Q3: What is the regulatory pathway?

UNKNOWN. Not yet determined. Requires pre-submission to FDA. No predicate identified for some packages.

Q4: Who owns the IP?

UNKNOWN. IP ownership not verified. Requires legal counsel engagement.

Q5: What would it cost to validate?

First experiment: first physical article implementing the proposed intervention (GEOMETRY ABSENT — must be designed). Estimated effort from build plan. Full validation cost: COST TO BE QUOTED.

Commercial Interest

TARGET APPLICATION: CSF shunt systems for hydrocephalus management

CURRENT ALTERNATIVE: Standard fixed-pressure or programmable CSF shunt valves

MECHANISM DIFFERENCE: smart battery management systems that enhance resilience while maintaining energy density

POTENTIAL VALUE DRIVER: Addresses: Documented recurring problem 'lithium-ion consumer-product battery fire': 5 of 50 CPSC recalls in the retrieved window involve batteries/chargers/power electronics. Recalled products: 'Cooluli Recalls 10-Liter and 15-Liter Minifridges Due to Fire and Burn Hazards'; 'Shengsen Prank Sound Makers Recalled Due to Risk of Serious Injury or Death from Battery Ingestion'; 'A2batt Recalls EEMB Lithium Coin Battery Chargers Due to Risk of Serious Injury or Death from Batter'. LIMITS STAMPED INLINE: a recall is an acknowledged product hazard, NOT an incidence rate; injury lists are associated complaints, not a census (Art. XXI.5 analog).

ADOPTION BARRIER: Requires physical validation, regulatory clearance, and manufacturing qualification

DEVELOPMENT BARRIER: No physical prototype exists; manufacturing process not established

Licensee Capability Fit

IDEAL BUYER: domain-appropriate technology buyer (to be identified by portfolio owner)

REQUIRED ENGINEERING CAPABILITY: Catheter design, fluid mechanics / domain-specific engineering

REQUIRED MANUFACTURING CAPABILITY: Medical tubing extrusion, injection molding, cleanroom assembly

REQUIRED REGULATORY CAPABILITY: FDA 510(k) or PMA submission experience

REQUIRED CAPITAL: Validation: \$50-200K. Full development: \$500K-2M (estimates, not quoted)

REQUIRED MARKET ACCESS: Neurosurgery market channels; hospital purchasing relationships

1. Technology Description

Domain: mechanical_structural

Disciplines: mechanical engineering, structural analysis, fatigue and fracture, materials science

Architecture: PROPOSED architecture (no physical system exists): load path / structural backbone; actuation or compliance element; guidance / bearing interfaces; fatigue-critical features register — realizing: Implementing a cost-effective battery management system with enhanced monitoring and control capabilities that can detect abnormal charging patterns, prevent overcharging, and isolate cells showing signs of degradation

Subsystems:

?: load path / structural backbone - ?

?: actuation or compliance element - ?

?: guidance / bearing interfaces - ?

?: fatigue-critical features register - ?

2. Mechanism Architecture

Physical Changes: PROPOSED: Implementing a cost-effective battery management system with enhanced monitoring and control capabilities that can detect abnormal charging patterns, prevent overcharging, and isolate cells showing signs of degradation

Key Physics: Mechanics / structures governing relations proposed for this invention; SYMBOLIC ONLY until inputs are sourced

3. Governing Engineering Model

Summary: Mechanics / structures governing relations proposed for this invention; SYMBOLIC ONLY until inputs are sourced

Equations:

MEC-002: Section bending stress. Expression: $\sigma = M * c / I$. Applicability to this invention: CONDITIONAL; holds under: linear-elastic material; small deflection; prismatic section. Assumptions: stress concentrators multiply local stress above the nominal value — geometry factors must be applied from the actual design. Engaged variables: NOT ESTABLISHED.

MEC-003: Hooke's law (linear elasticity). Expression: $\sigma = E * \epsilon$. Applicability to this invention: CONDITIONAL; holds under: below proportional limit; isothermal. Assumptions: superelastic members (e.g. nitinol) do not follow linear elasticity through transformation — use their measured plateau relations instead. Engaged variables: NOT ESTABLISHED.

MEC-001: Euler buckling load. Expression: $P_{cr} = \pi^2 * E * I / (K * L)^2$. Applicability to this invention: CONDITIONAL; holds under: elastic, slender column, small deflection. Assumptions: K for catheter constraint conditions is MODEL_DERIVED. Engaged variables: NOT ESTABLISHED.

Assumptions:

- stress concentrators multiply local stress above the nominal value — geometry factors must be applied from the actual design
- superelastic members (e.g. nitinol) do not follow linear elasticity through transformation — use their measured plateau relations instead
- K for catheter constraint conditions is MODEL_DERIVED
- MEC-002 (Section bending stress) is applicable only under: linear-elastic material; small deflection; prismatic section.
- MEC-003 (Hooke's law (linear elasticity)) is applicable only under: below proportional limit; isothermal.
- MEC-001 (Euler buckling load) is applicable only under: elastic, slender column, small deflection.
- Model input variables requiring sourced values before any numeric result: M (N.m), c (m), I (m⁴), E (Pa), epsilon (-), E (Pa), I (m⁴), K (-), L (m) — all UNKNOWN (no sourced value exists; Art. VIII).
- Model output variables produced by the governing model: sigma (Pa), sigma (Pa), P_cr (N) — UNKNOWN until the inputs are sourced and the model is evaluated.

- Critical parameters under specification (parameters with recorded units and epistemic status; no values invented): member slenderness ratio [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; yield/ultimate strength [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; applied cyclic load amplitude [P_cr, N] value_status=UNKNOWN; required run-out cycles [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; articulation radius [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN; friction coefficient at guides [NOT_ASSIGNED, UNKNOWN] value_status=UNKNOWN.

Boundary Conditions:

- to be established by the first bench characterization (not yet performed)

Failure Regimes:

- {'regime': 'battery fire initiation in consumer products', 'basis': 'SOURCE_FACT (problem.json)'}
- {'regime': 'kink/buckling below required push ability', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'fatigue crack initiation at stress concentrator', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'wear debris generation at sliding interfaces', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'joint separation under overload', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}
- {'regime': 'set-release of superelastic shape after set', 'basis': 'domain registry candidate from the engineering domain registry — applicability to THIS invention recorded below'}

4. Design Inputs

ID	Input	Value	Origin
DI-001	Stated constraint	Prevention must work across consumer misuse scenarios (overcharge, mechanical damage, cheap chargers) at consumer level	SOULACI_FAGT
DI-002	Mechanism driver (proposed)	smart battery management systems that enhance resilience while maintaining energy density	MODIFIED
DI-003	Expected effect target	Reduction in lithium-ion battery fires by preventing thermal runaway through early detection and intervention, even in cases of mechanical damage	MODIFIED
DI-004	Domain design input pattern: delivery path applicability	NOT_ESTABLISHED (candidate pattern; confirm against engineering profile)	ENGINEERING_PROPOSED
DI-005	Domain design input pattern: actuation or compliance	NOT_ESTABLISHED (candidate pattern; confirm against engineering profile)	ENGINEERING_PROPOSED
DI-006	Domain design input pattern: duty cycle applicability	NOT_ESTABLISHED (candidate pattern; confirm against engineering profile)	ENGINEERING_PROPOSED
DI-007	Regulatory design input: compliance standards	ASTM F2516 (lithium-ion battery safety) pathway UNKNOWN (NEITHER PROPOSED)	NEITHER PROPOSED
DI-008	Custodied observation: A Future Perspective on Wearable Battery	https://www.sciencedirect.com/science/article/pii/S0924646018300442; content_hash 36498242; lochash 163d11e7909b0e810512433e0dae54afd9f7217f21b4ac	https://www.sciencedirect.com/science/article/pii/S0924646018300442; lochash 163d11e7909b0e810512433e0dae54afd9f7217f21b4ac
DI-009	Custodied observation: The Materials Science Behind Sustainable Batteries	https://behindthesciences.com/2018/05/14/sustainable-batteries/; content_hash 1e8ac48211f659219e5c9950199887bf60a05c703159f6915	https://behindthesciences.com/2018/05/14/sustainable-batteries/; content_hash 1e8ac48211f659219e5c9950199887bf60a05c703159f6915
DI-010	Custodied observation: From Present to Future: Portable Power	https://www.energypower.com/2018/05/14/portable-power/; content_hash 5820b39b0cb574f4e3f75324b532df076abbc94ff3478	https://www.energypower.com/2018/05/14/portable-power/; content_hash 5820b39b0cb574f4e3f75324b532df076abbc94ff3478
DI-011	Custodied observation: Ensuring Battery Safety in Electric Vehicles	https://www.electrive.com/2018/05/14/ensuring-battery-safety-in-electric-vehicles/; content_hash 3a35fae1e380b5f064eac636e831a35b1b747ed2c6f	https://www.electrive.com/2018/05/14/ensuring-battery-safety-in-electric-vehicles/; content_hash 3a35fae1e380b5f064eac636e831a35b1b747ed2c6f
DI-012	Custodied observation: Microgrid as a Tool for Energy Self-Sufficiency	https://www.energypower.com/2018/05/14/microgrid-as-a-tool-for-energy-self-sufficiency/; content_hash 4a4d404ce9c28200e27e54a5191c19a636c2978396946	https://www.energypower.com/2018/05/14/microgrid-as-a-tool-for-energy-self-sufficiency/; content_hash 4a4d404ce9c28200e27e54a5191c19a636c2978396946

5. Design Outputs

ID	Description	Status	Missing Inputs
DO-001	load path / structural backbone realizing Implementing a cost-effective battery management system with enhanced monitoring and control capabilities that	CONCEPTUAL	designer-art system definition
DO-002	actuation or compliance element realizing Implementing a cost-effective battery management system with enhanced monitoring and control capabilities that	CONCEPTUAL	designer-art system definition
DO-003	guidance / bearing interfaces realizing Implementing a cost-effective battery management system with enhanced monitoring and control capabilities that	CONCEPTUAL	designer-art system definition
DO-004	fatigue-critical features register realizing Implementing a cost-effective battery management system with enhanced monitoring and control capabilities that	CONCEPTUAL	designer-art system definition
DO-005	load path / structural backbone interfaces with actuation or compliance element: signal/energy/artificial intelligence (diffusions, tolerances)	CONCEPTUAL	signal/energy/artificial intelligence (diffusions, tolerances)
DO-006	actuation or compliance element interfaces with guidance / bearing interfaces: signal/energy/artificial intelligence (diffusions, tolerances)	CONCEPTUAL	signal/energy/artificial intelligence (diffusions, tolerances)
DO-007	guidance / bearing interfaces interfaces with fatigue-critical features register: signal/energy/artificial intelligence (diffusions, tolerances)	CONCEPTUAL	signal/energy/artificial intelligence (diffusions, tolerances)
DO-008	Device boundaries: patient/fluid inlet and outlet terminal control/data points required by telemedicine segmentation effective battery management	CONCEPTUAL	control/data points required by telemedicine segmentation
DO-009	Flow/force/signal path geometry realizing the mechanism is unknown Implementing a cost-effective battery management system with enhanced monitoring and control	UNKNOWN	Implementing a cost-effective battery management system with enhanced monitoring and control
DO-010	Operating range for member slenderness ratio within which the mechanism holds its stated behavior	UNKNOWN	mechanism holds its stated behavior
DO-011	Operating range for yield/ultimate strength within which the mechanism holds its stated behavior	UNKNOWN	mechanism holds its stated behavior
DO-012	Operating range for applied cyclic load amplitude within which the mechanism holds its stated behavior	UNKNOWN	mechanism holds its stated behavior
DO-013	Operating range for required run-out cycles within which the mechanism holds its stated behavior	UNKNOWN	mechanism holds its stated behavior
DO-014	Operating range for articulation radius within which the mechanism holds its stated behavior	UNKNOWN	mechanism holds its stated behavior
DO-015	Operating range for friction coefficient at guides within which the mechanism holds its stated behavior	UNKNOWN	mechanism holds its stated behavior
DO-016	Sensing -> decision -> actuation loop maintaining the PROPOSED operating point in a predicted trajectory; effective battery management (segment values required)	PROPOSED	operating point in a predicted trajectory; effective battery management (segment values required)
DO-017	Information path: sensing element -> decision/monitor or CONCEPtual act (minimal); rich data paths are NOT APPLICABLE to this mechanical/physical in	CONCEPTUAL	act (minimal); rich data paths are NOT APPLICABLE to this mechanical/physical in
DO-018	Fixture enabling: buckling / kink margin tests across duty cycle/rig concept named, build specific instrumentation selection; acceptance instrumentation	UNKNOWN	rig concept named, build specific instrumentation selection; acceptance instrumentation
DO-019	Fixture enabling: fatigue life (run-out) testing — rig concept named, build specific instrumentation selection; acceptance instrumentation	UNKNOWN	rig concept named, build specific instrumentation selection; acceptance instrumentation
DO-020	Fixture enabling: friction and wear characterization — rig concept named, build specific instrumentation selection; acceptance instrumentation	UNKNOWN	rig concept named, build specific instrumentation selection; acceptance instrumentation

ID	Description	Status	Missing Inputs
DO-021	Fixture enabling: buckling/kink margin test across duty cycle	UNKNOWN	Unknown concept name; fixture specification; instrument selection; acceptance instrumentation

6. Critical Design Parameters

Name	Value	Unit	Basis	Verification
member slenderness ratio	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method
yield/ultimate strength	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method
applied cyclic load amplitude	UNKNOWN (no sourced value)	N	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method
required run-out cycles	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method
articulation radius	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method
friction coefficient at guides	UNKNOWN (no sourced value)	UNKNOWN	ENGINEERING_PROPOSED	(design assigned candidate) value and verification method

7. Failure Modes

Mode	Mechanism	Mitigation	Verification	Residual Uncertainty
battery fire initiation in failure mode	Failure by fire propagation	designs cover by NOTC-STB-94-02 (requires design assigned candidate)	UNKNOWN	Non-triggered outside the operating cycle (Attrib)
kink/buckling below required	Failure by kink/buckling below required	compressions below ultimate capacity and supports capacity (design assigned candidate)	UNKNOWN	Not tested electrically fatigue load due (Attrib)
fatigue crack initiation in failure mode	Failure by fatigue crack initiation	compressions below fatigue limit; shear below shear limit; lap joints below lap joint limit	UNKNOWN	Not tested electrically fatigue load due (Attrib)
wear debris generation in failure mode	Failure by wear debris generation	material painting and coating of contact pressure reduction	UNKNOWN	Not tested electrically fatigue load due (Attrib)
joint separation under failure mode	Failure by joint separation under failure mode	determine load and size of joint	UNKNOWN	Not tested electrically fatigue load due (Attrib)
set-release of superelastic	Failure by set-release of superelastic	shape shape of set-release	UNKNOWN	Not tested electrically fatigue load due (Attrib)

8. Failure Analysis

Failure Mode	Mechanism	Mitigation	Residual Uncertainty
battery fire initiation in consumer products	Failure by battery fire initiation in consumer products	design controls, the NOT ESTABLISHED	residual uncertainty is low, no significant change in the operating (Annex V)
kink/buckling below required capacity	Failure by kink/buckling below required capacity	pushability test, bendup test, and kinkdown test, the residual uncertainty is low, no significant change in the operating (Annex V)	
fatigue crack initiation at stress concentration	Failure by fatigue crack initiation at stress concentration	concentration of stress at surfaces, the residual uncertainty is low, no significant change in the operating (Annex V)	
wear debris generation at sliding interfaces	Failure by wear debris generation at sliding interfaces	wear debris generation, the residual uncertainty is low, no significant change in the operating (Annex V)	
joint separation under overload	Failure by joint separation under overload	load on a joint, the residual uncertainty is low, no significant change in the operating (Annex V)	
set-release of superelastic shape memory alloy	Failure by set-release of superelastic shape memory alloy	shape memory alloy, the residual uncertainty is low, no significant change in the operating (Annex V)	

9. Verification Strategy

ID	Requirement	Method	Acceptance	Result
VF-001	Comparative testing of battery packs with different battery chemistries	Comparative testing of battery packs with different battery chemistries	the residual uncertainty is low, no significant change in the operating (Annex V)	the residual uncertainty is low, no significant change in the operating (Annex V)
VF-002	run killer experiment: falsification_test	run killer experiment: falsification_test	the residual uncertainty is low, no significant change in the operating (Annex V)	the residual uncertainty is low, no significant change in the operating (Annex V)
VF-003	buckling/kink margin test across the duty cycle	buckling/kink margin test across the duty cycle	the residual uncertainty is low, no significant change in the operating (Annex V)	the residual uncertainty is low, no significant change in the operating (Annex V)
VF-004	fatigue run-out at worst-case loading	fatigue run-out at worst-case loading	the residual uncertainty is low, no significant change in the operating (Annex V)	the residual uncertainty is low, no significant change in the operating (Annex V)
VF-005	buckling / kink margin tests across duty cycle	buckling / kink margin tests across duty cycle	the residual uncertainty is low, no significant change in the operating (Annex V)	the residual uncertainty is low, no significant change in the operating (Annex V)
VF-006	overload margin demonstration to worst-case	overload margin demonstration to worst-case	the residual uncertainty is low, no significant change in the operating (Annex V)	the residual uncertainty is low, no significant change in the operating (Annex V)

10. Validation Strategy

ID	Requirement	Method	Acceptance	Result
VA-001	end-use validation of verified design	NOT ESTABLISHED (defined after verification)	the residual uncertainty is low, no significant change in the operating (Annex V)	Validation is required; validation is required

11. Materials

Component	Candidate Material	Source	Verification Required	Status
domain candidate material	nitinol / polymer composite membrane	minimally invasive device practice	True	ENGINEERING_PROPOSED

12. Bill of Materials

Item	Description	Qty	Type	Material	Supplier	Criticality	Verification
nitinol / polymer composite membrane	nitinol / polymer composite membrane	1	Material	nitinol / polymer composite membrane	NOT ESTABLISHED	NOT ESTABLISHED	Reliability verification required

13. Manufacturing

Status: NO MANUFACTURING ANALYSIS POSSIBLE — no design exists; processes are candidates only

Candidate Processes:

- laser-cut hypotube / braided composite shaft: domain registry manufacturing route - candidate only; no process qualification exists
- passivation and surface finish control: domain registry manufacturing route - candidate only; no process qualification exists

14. External Evidence

Source 1:

URL :

Excerpt:

15. Transfer Boundary

YOU RECEIVE:

- + Buyer receives: canonical invention specification (with epistemic classes).
- + Buyer receives: engineering specification (this document, all blocks classed).
- + Buyer receives: design input/output graph with explicit IDs.
- + Buyer receives: engineering reasoning chains (claim->principle->model->input->assumption->output->failure->verification).
- + Buyer receives: killer-experiment definition and its EIG basis.
- + Buyer receives: evidence custody references and hashes.
- + Buyer receives: unresolved-uncertainty register.
- + Buyer receives: domain equation set (symbolic; sources and applicability judgments identified).

YOU MUST DEVELOP:

- > Buyer must create: actual geometry and detailed design.
- > Buyer must create: sourced engineering parameters and acceptance thresholds.
- > Buyer must create: physical prototypes.
- > Buyer must create: all verification and validation results.
- > Buyer must create: manufacturing process qualification.
- > Buyer must create: regulatory pathway determination and submissions.

YOU MUST VERIFY:

- ? Biocompatibility (ISO 10993)
- ? Sterilization compatibility
- ? Mechanical integrity
- ? Regulatory pathway
- ? IP ownership and freedom to operate
- ? Manufacturing process capability

16. Open Questions

- {'unknown': 'FM_WITHOUT_QUANTITY_MATCHED_VERIFICATION:FM-DOM-005', 'reason': 'no available verification method measures a quantity family this failure affects (E21-A quantity rule); UNKNOWN disclosure recorded on the row'}
- {'unknown': 'value of critical parameter member slenderness ratio [NOT_ASSIGNED UNKNOWN]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-001'}
- {'unknown': 'value of critical parameter yield/ultimate strength [NOT_ASSIGNED UNKNOWN]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-002'}
- {'unknown': 'value of critical parameter applied cyclic load amplitude [P_cr N]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-003'}
- {'unknown': 'value of critical parameter required run-out cycles [NOT_ASSIGNED UNKNOWN]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-004'}
- {'unknown': 'value of critical parameter articulation radius [NOT_ASSIGNED UNKNOWN]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-005'}
- {'unknown': 'value of critical parameter friction coefficient at guides [NOT_ASSIGNED UNKNOWN]', 'reason': 'no sourced value exists (value_status UNKNOWN); required before any numeric design decision', 'binding': 'CP-006'}
- {'unknown': 'acceptance criterion for verification VF-001', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-001'}
- {'unknown': 'acceptance criterion for verification VF-002', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-002'}

- {'unknown': 'acceptance criterion for verification VF-003', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-003'}
- {'unknown': 'acceptance criterion for verification VF-004', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-004'}
- {'unknown': 'acceptance criterion for verification VF-005', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-005'}
- {'unknown': 'acceptance criterion for verification VF-006', 'reason': 'criterion status ENGINEERING_PROPOSED: no sourced threshold exists; the numeric margin is fixed at pre-registration from the measured baseline', 'binding': 'VF-006'}

17. Decisive Next Experiment

WP	Test Article	Equipment	Measurement	Acceptance	Deliverable	Effort
WP-01	first physical article	implementing the proposed physical verification (to be completed by SEP-2023)	physical verification (to be completed by SEP-2023)	NOT ESTABLISHED (to be quoted)	Article 103-113 (to be quoted)	135 (to be quoted)
WP-02	constraint-compliance envelope article	per method	constraint-compliance envelope article	NOT ESTABLISHED (to be quoted)	Article 103-113 (to be quoted)	135 (to be quoted)
WP-03	article for VF-001	per method	Comparative testing of VF-001	NOT ESTABLISHED (to be quoted)	Article 103-113 (to be quoted)	135 (to be quoted)
WP-04	article for VF-002	per method	run killer experiment: VF-002	NOT ESTABLISHED (to be quoted)	Article 103-113 (to be quoted)	135 (to be quoted)
WP-05	article for VF-003	per method	buckling/kink margin test: VF-003	NOT ESTABLISHED (to be quoted)	Article 103-113 (to be quoted)	135 (to be quoted)
WP-06	article for VF-004	per method	fatigue run-out at work: VF-004	NOT ESTABLISHED (to be quoted)	Article 103-113 (to be quoted)	135 (to be quoted)
WP-07	article for VF-005	per method	buckling / kink margin test: VF-005	NOT ESTABLISHED (to be quoted)	Article 103-113 (to be quoted)	135 (to be quoted)
WP-08	article for VF-006	per method	overload margin demonstration: VF-006	NOT ESTABLISHED (to be quoted)	Article 103-113 (to be quoted)	135 (to be quoted)

18. Buyer Decision Framework

WHY INVEST: Engineering definition exists with 3 governing equations and a 8-step build plan. The next experiment is clearly specified. Target buyer: domain-appropriate technology buyer (to be identified by portfolio owner).

WHY WAIT: No physical prototype exists. Regulatory pathway is unresolved. IP ownership is unverified. Manufacturing process is not qualified.

WHY REJECT: killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.44573539382492045, EIG-per-cost 0.44573539382492045; the falsification test carries the pre-registered pass/fail before any transfer)

WHAT WOULD CHANGE THE DECISION: A successful bench experiment demonstrating the core mechanism. A qualified regulatory pathway. A confirmed IP position.

19. Kill Condition

KILL_IF: killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.44573539382492045, EIG-per-cost 0.44573539382492045; the falsification test carries the pre-registered pass/fail before any transfer)

DISCLOSURE

This dossier distinguishes established evidence, external precedent, computational/modelled results, engineering proposals, and unresolved questions. The absence of physical validation should not be interpreted as evidence that the technology will fail; it means the relevant proposition has not yet been established experimentally.

This dossier does not constitute legal, patentability, FTO, regulatory, or investment advice.